Model Analysis Report

Spatial Interpolation Model

MKM-SP-001

Report Date: 27 March 2026

Governance Framework: SR 11-7 / SS1/23 Model Risk Management

Contents

1 Sensitivity Analysis

This section presents the sensitivity of model outputs to key input parameters. Parameter perturbation ranges are chosen to cover plausible operating conditions and stress scenarios as required under SR 11-7 guidance.

Table 1: Interpolated WSE sensitivity to IDW power parameter (3 gauges).

Power p	**WSE (m)**
0.5000	11.9
1.00	12.2
1.50	12.4
2.00	12.4
2.50	12.5
3.00	12.5
4.00	12.5

Table 2: Interpolated WSE sensitivity to nearest gauge distance ($p = 2$).

Nearest gauge (m)	**WSE (m)**
100	12.5
250	12.5
500	12.4
1000	12.3
2000	12.0
5000	11.7
10000	11.6

Table 3: Interpolated WSE sensitivity to number of contributing gauges ($p = 2$).

Gauges used	**WSE (m)**
1	12.5
2	12.4
3	12.4
4	12.4
5	12.4

2 Stress Testing

This section documents stress testing scenarios applied to the model under extreme but plausible conditions.

2.1 Stress Scenarios

Table 4: Stress testing scenarios for Spatial Interpolation Model.

Scenario	Parameter Shock	Severity	Status
Baseline	No shock	—	Passed
Moderate stress	$+2\sigma$ inputs	Medium	Pending
Severe stress	$+3\sigma$ inputs	High	Pending
Reverse stress	Output threshold breach	Critical	Pending

Detailed stress test results will be populated as scenarios are executed. Refer to the model's validation plan for the full stress testing programme.

3 Model Performance

This section tracks key performance metrics for the model across validation and production environments.

Table 5: Performance metrics for Spatial Interpolation Model.

Metric	Value	Status
Unit test pass rate	See Test Results	—
Execution time (single run)	TBD	—
Numerical stability	TBD	—
Reproducibility	Deterministic	OK

This report is produced automatically; human review is required before formal model approval. Supporting artefacts are available in the model documentation directory.